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INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

Cell Cycle and cell Division

From unicellular to Multicellular

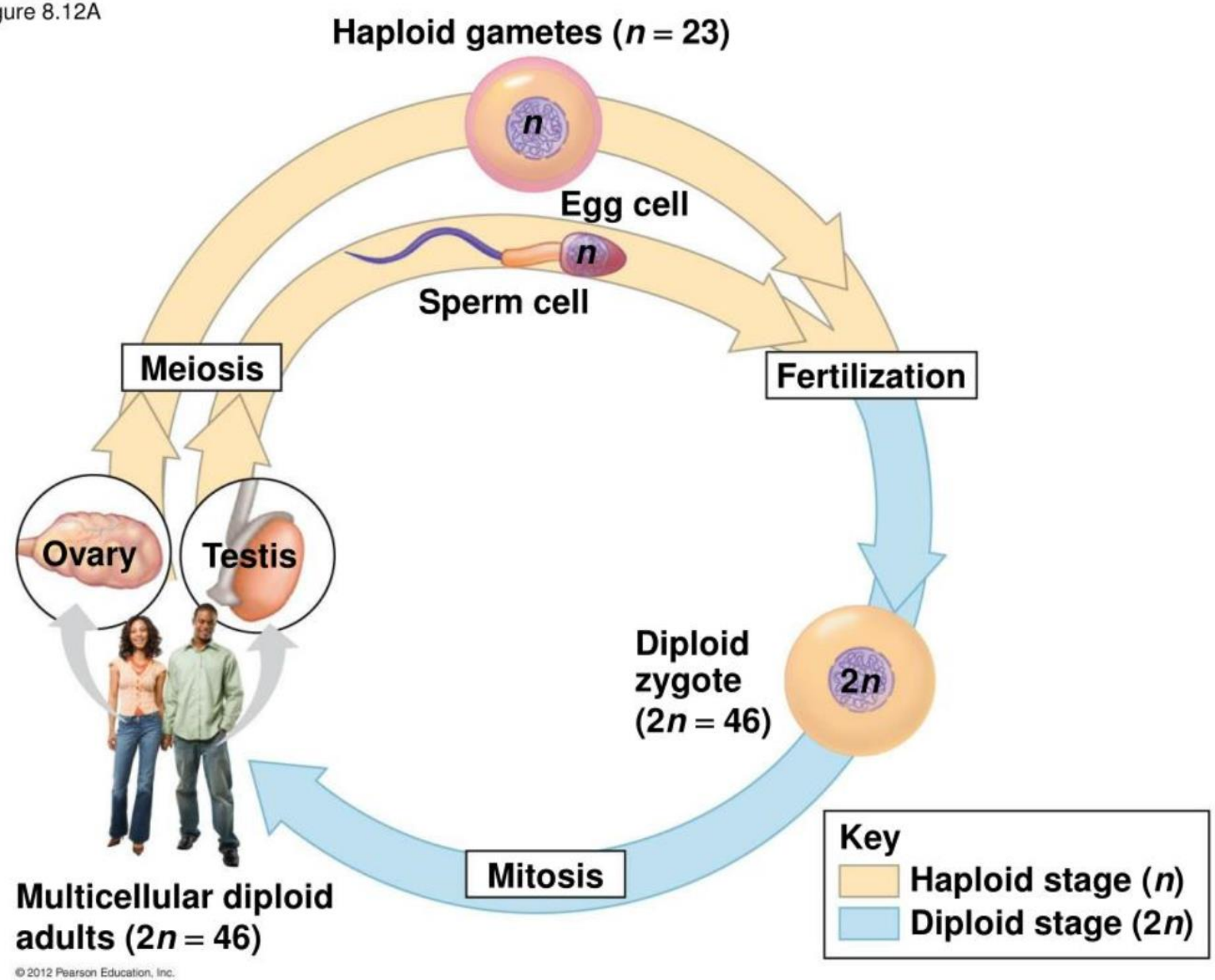
Lecture 20

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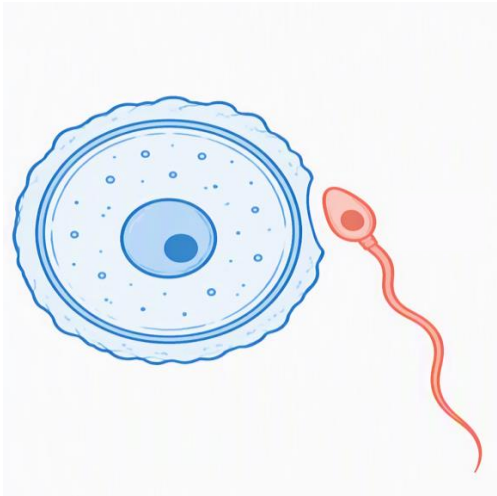
Figure 8.12A

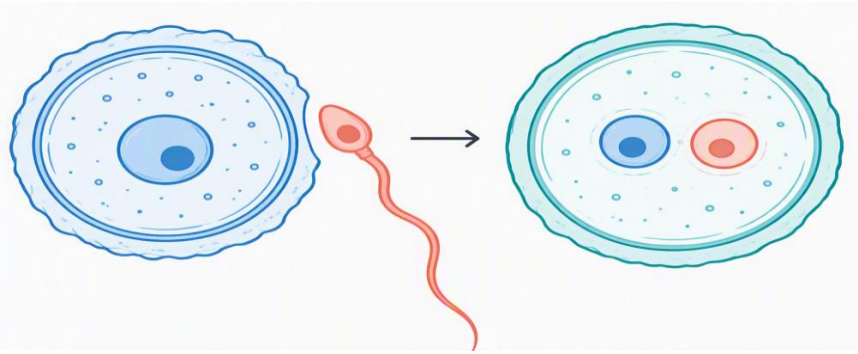


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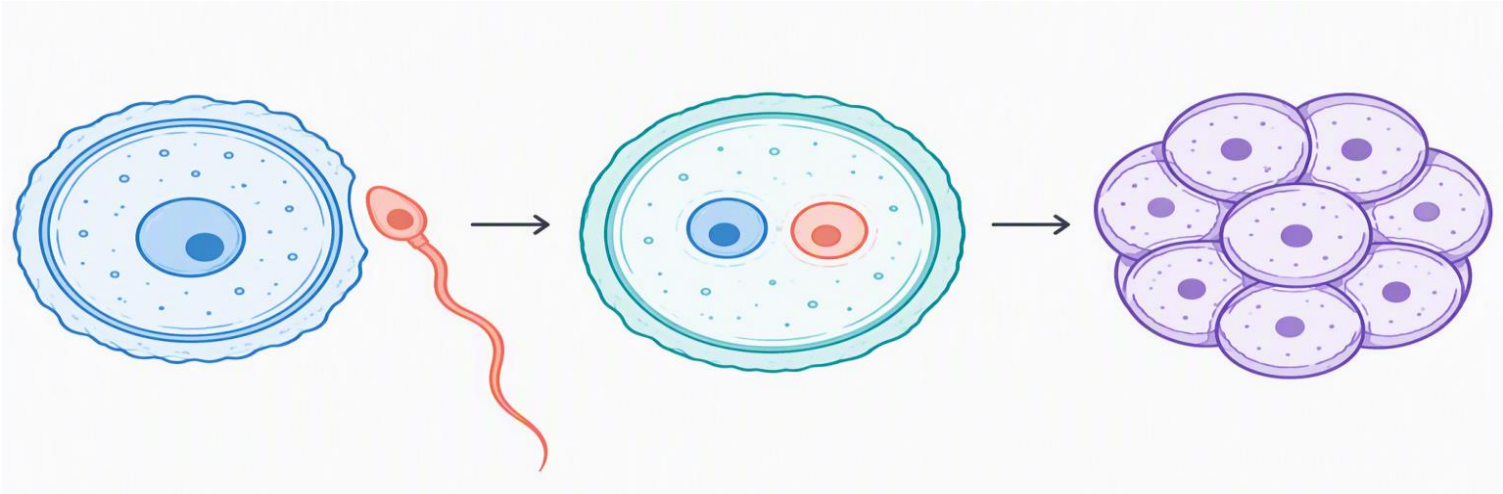
Few terminologies

- **Post Translational modification**: covalent and enzymatic or spontaneous changes that proteins undergo after synthesis (translation)
- **Gametes**: Reproductive cells
- **Somatic cells**: Body cells like muscle cells or nerve cells
- **Zygote**: Fertilized cell soon after the fusion of gametes
- **Haploid**: Cells containing only one chromosome from each pair
- **Diploid**: Contains the complete set of paired chromosomes
- **Mitosis**: Process of producing Diploid cells
- **Meiosis**: Process of producing Haploid cells





Gamete → Zygote → Embryo



GAMETES — Before fertilization

Specialised reproductive cells
Egg + sperm
Haploid: one chromosome set (n)

ZYGOTE — Immediately after fertilization

One fertilized cell
Formed by fusion of two gametes
Diploid: two chromosome sets (2n)

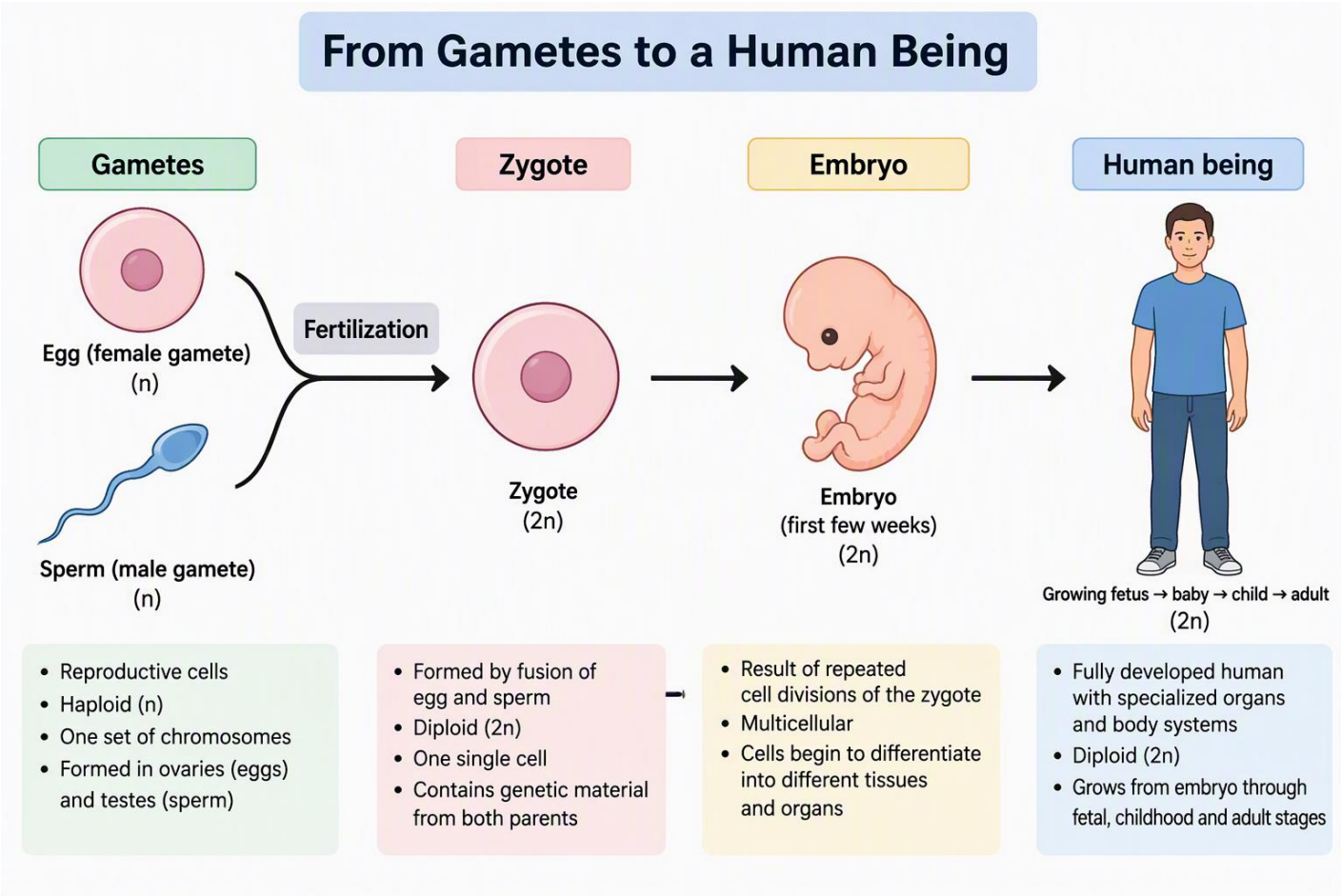
EMBRYO — After the zygote divides

Developing multicellular stage
Cells increase through mitosis
Begins differentiation and patterning

Key distinction: gametes are the cells that unite; the zygote is the first single cell formed; the embryo is the multicellular developing stage.

Cell cycle

- The cell cycle is the ordered series of growth and division steps a cell goes through to form two identical daughter cells.
- Growth and reproduction of all organisms depend on cell cycle
- Cell growth (cytoplasmic increase) is a continuous process.
  - Prepare itself for duplication – DNA replication, intracellular organelles
  - It's subsequent division to identical daughter cells – Cell division



Why does the cell cycle matter?

Growth

Increase cell number during development and tissue growth.

Repair & renewal

Replace worn-out or damaged cells in many tissues.

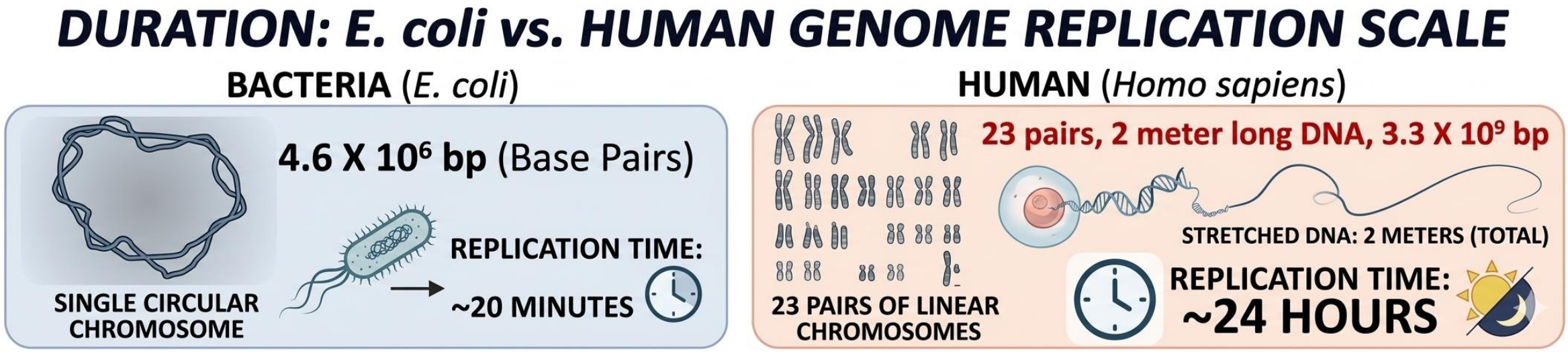
Reproduction

Cell division supports reproduction in many organisms.

One parent cell → two daughter cells

Cell types define cell cycle

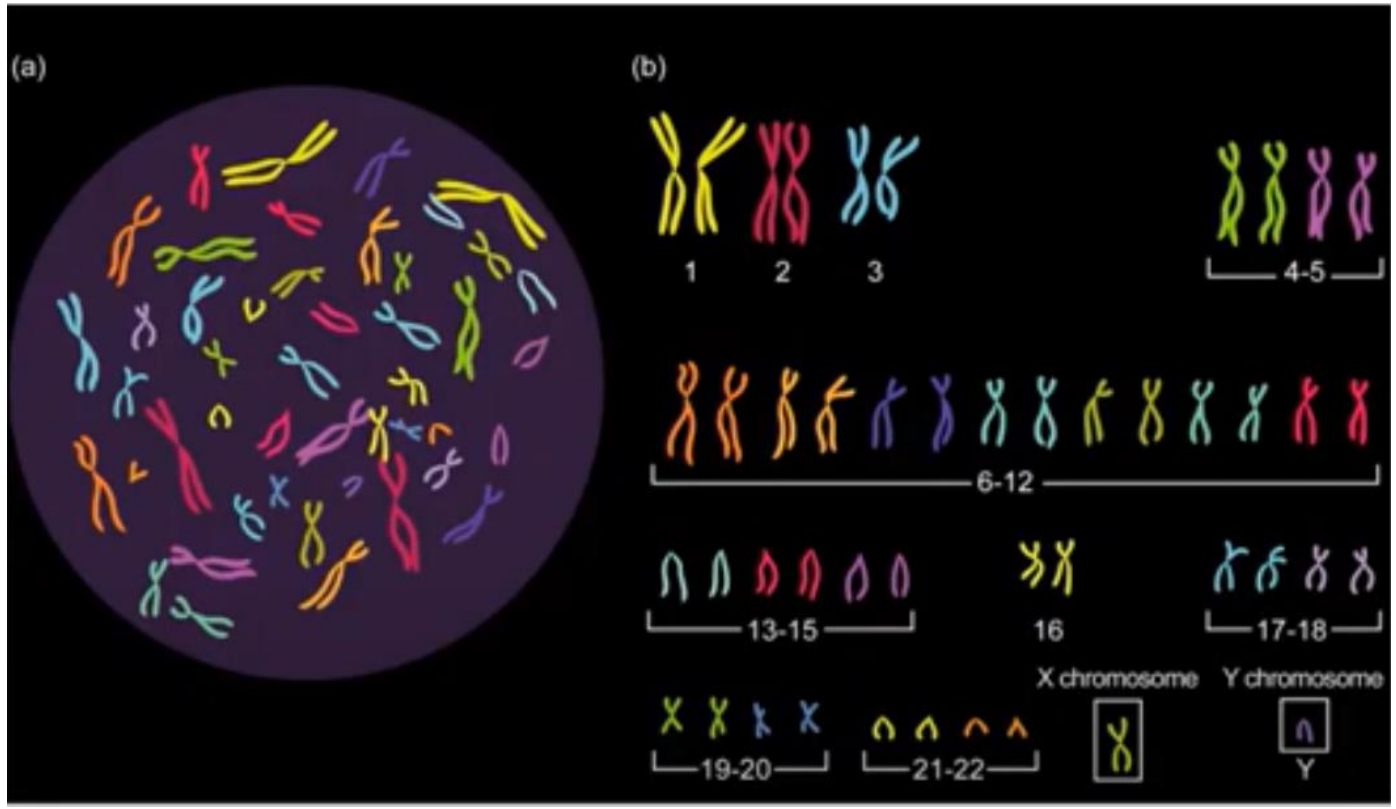
- Some specialized cells, such as nerve cell or RBCs, don't divide once formed
- Some cells divide on demand – liver cells, WBCs
- Most cells routinely divide in order to replenish the dead cells – skin cells, stem cells, gut cells



Duration of cell cycle varies according to cell types, but in eukaryotes it is typically 16-24 hours

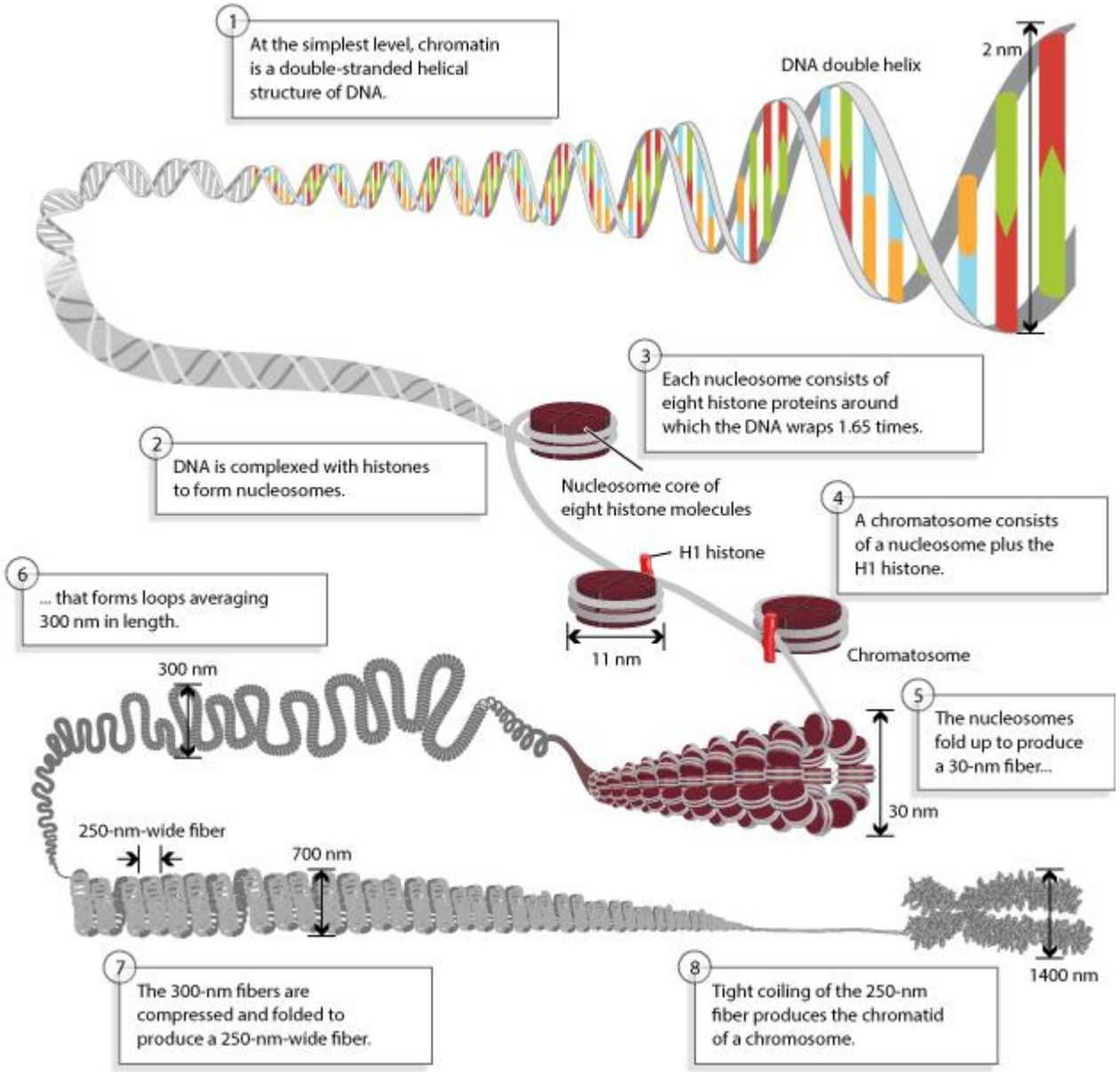
DNA Packaging

- A human cell contains 46 chromosomes
- Total DNA in a cell, end to end, is around 1.8m long
- Diameter of cell nucleus is approximately 2um
- $\frac{DNA\ length}{Nucleus\ Diameter} \approx 900{,}000!$



How does 1.8m DNA fit into 2um space!

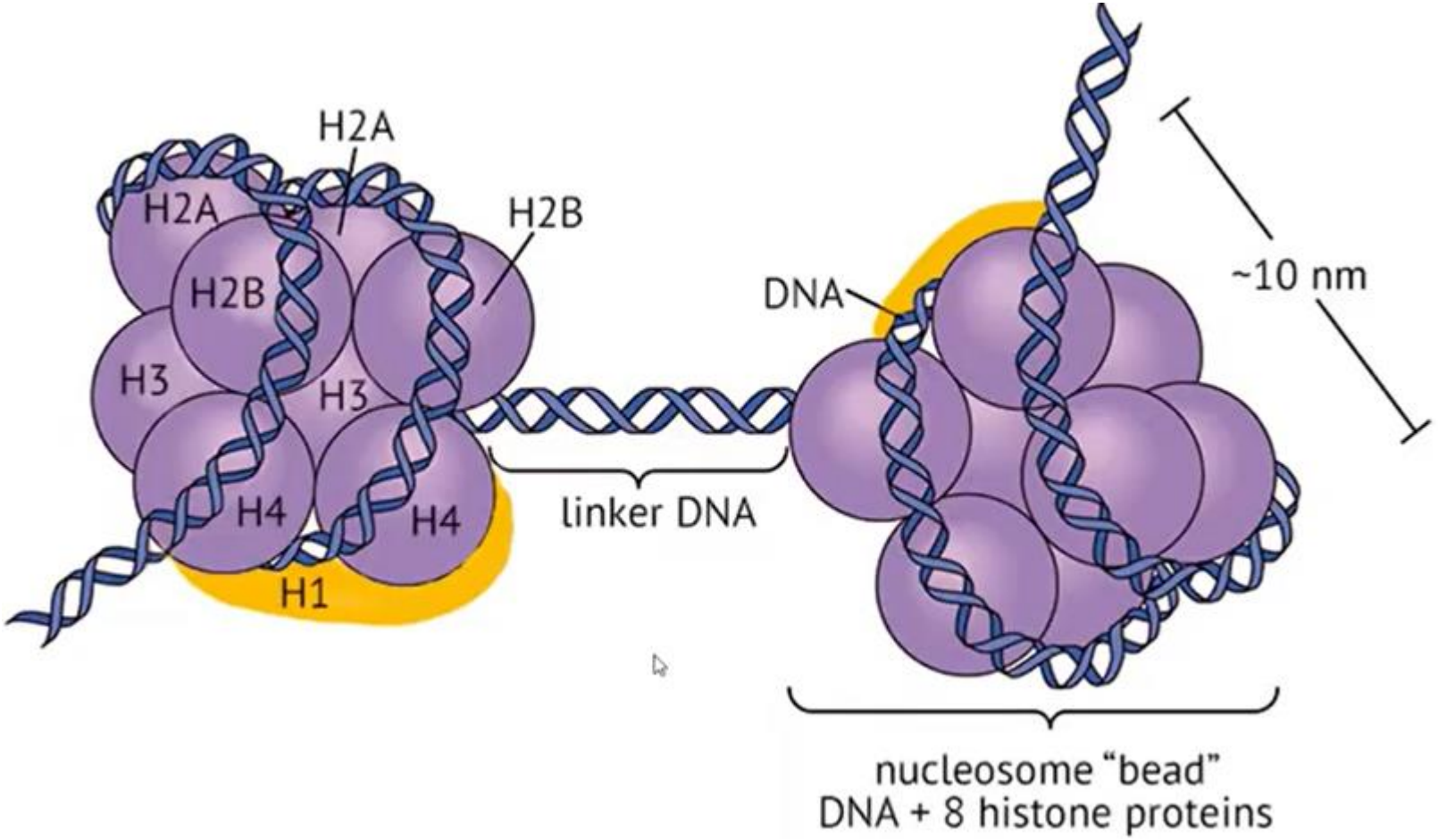
DNA and chromosomes



Histones are rich in Lysine and Arginine!

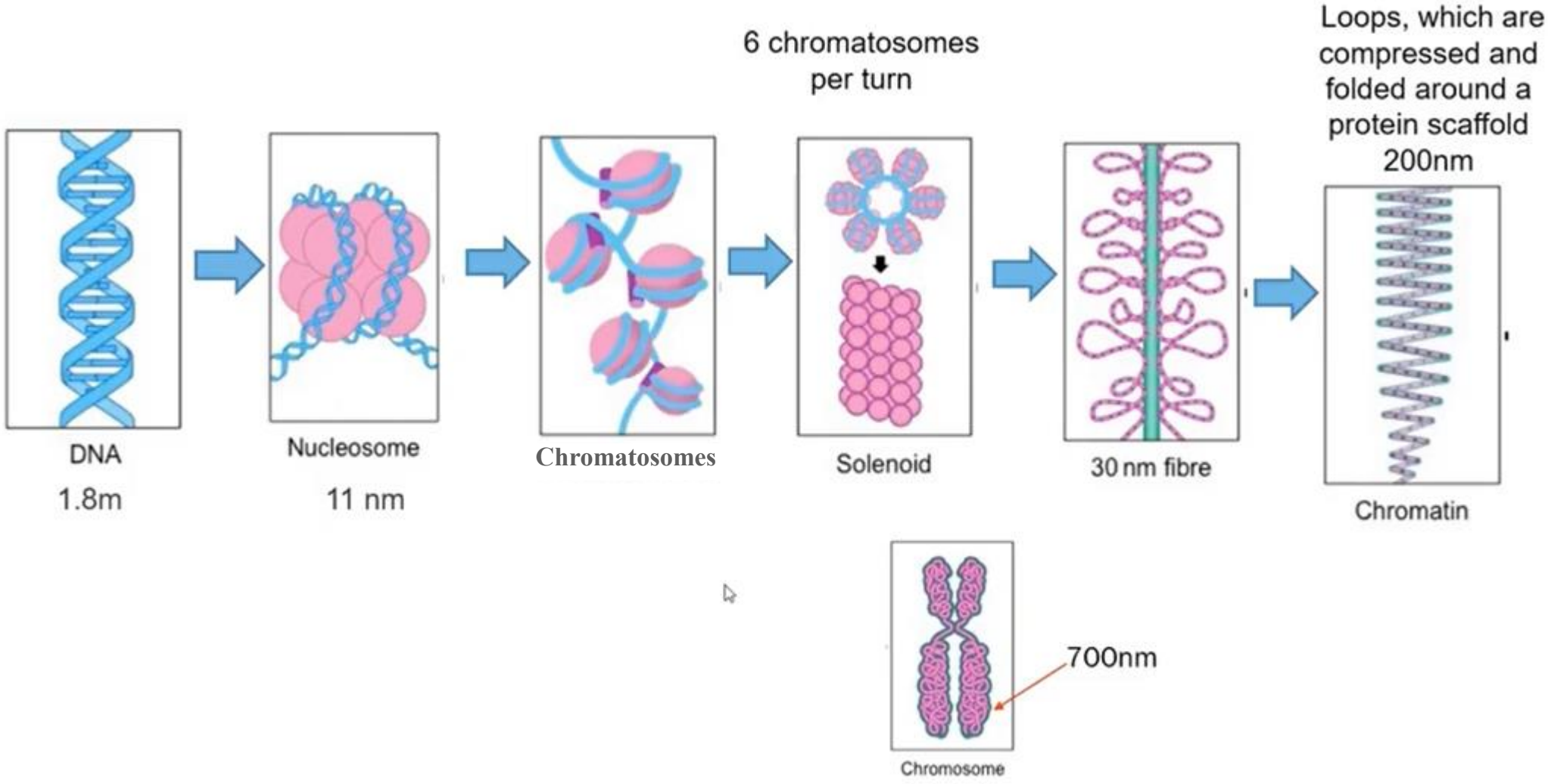
Annunziato, A. (2008) DNA Packaging: Nucleosomes and Chromatin. Nature Education 1(1):26

DNA and chromosomes



Histones has high content of lysine and arginine ⟶ +ve charge

DNA packaging steps

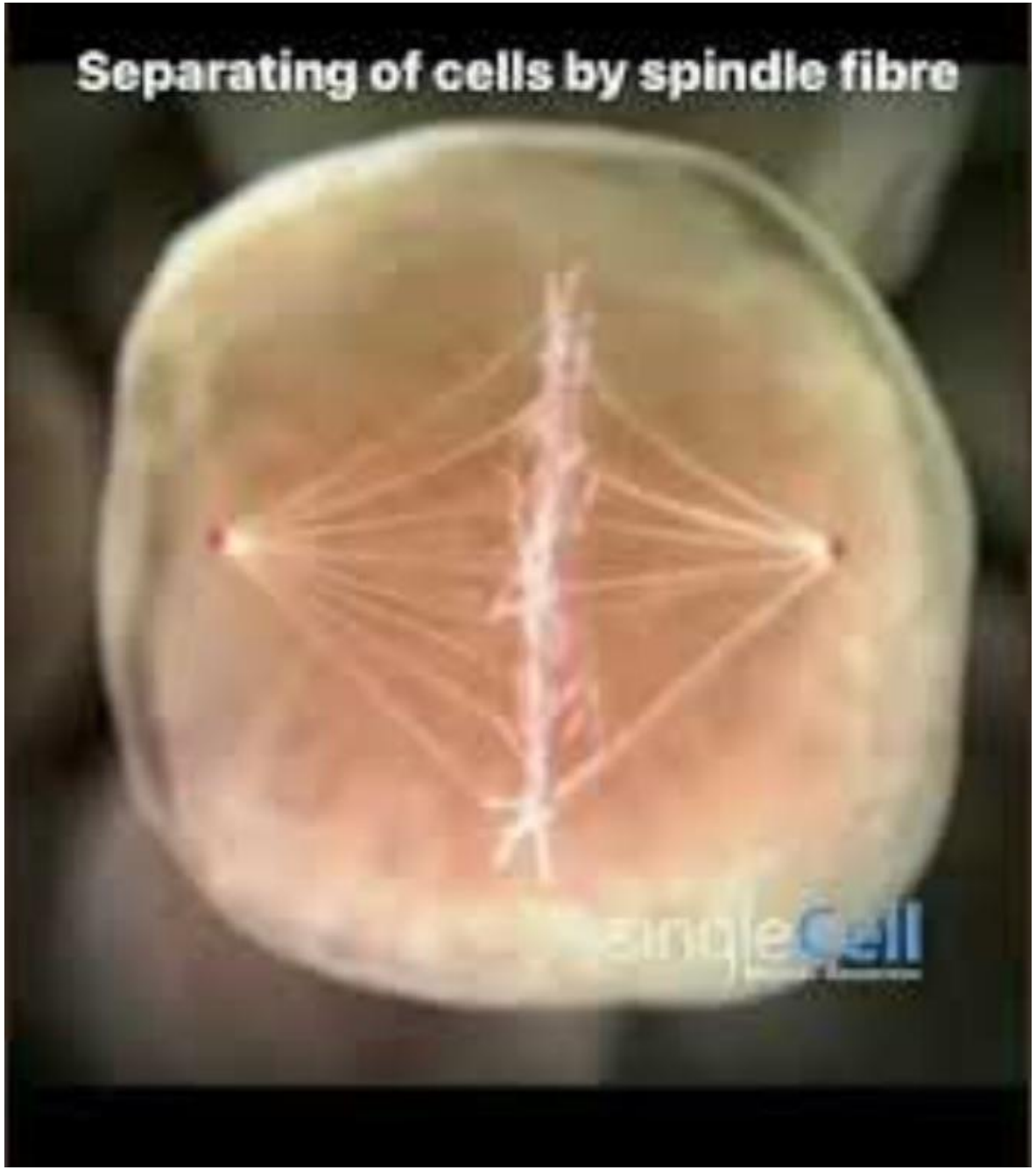


DNA packaging

- DNA is present inside our body in a highly compact form, where it goes under topological rearrangements such as supercoiling.

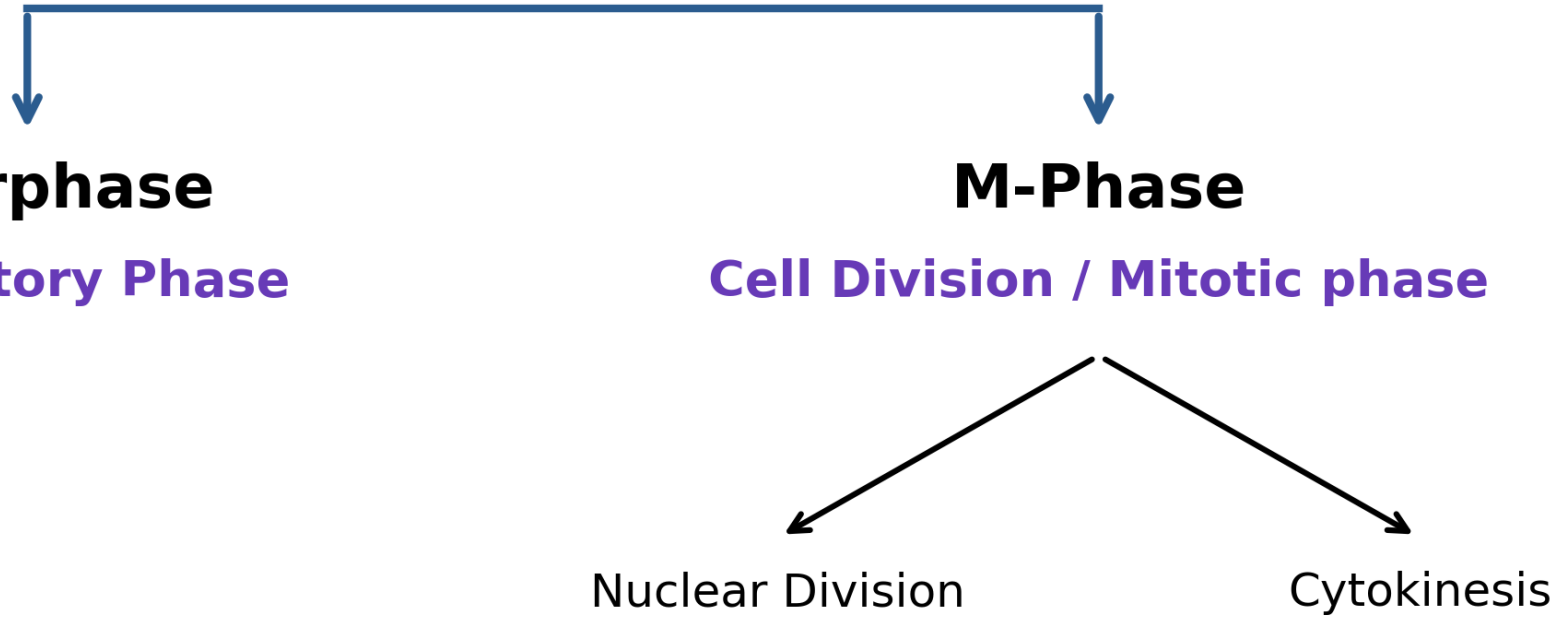


https://www.youtube.com/watch?v=gbSIBhFwQ4s



https://www.youtube.com/shorts/kn9iiwqFHTM

Cell Cycle



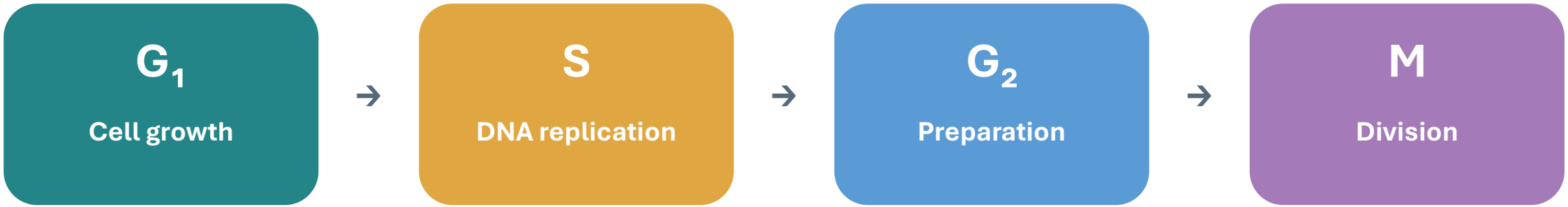
Interphase = G_1 + S + G_2

- G_1 - Cell Growth
- S - DNA replication
- G_2 - Preperation

Molecular Basis of ***cell cycle is conserved*** across all Eukaryotes

The cell cycle at a glance

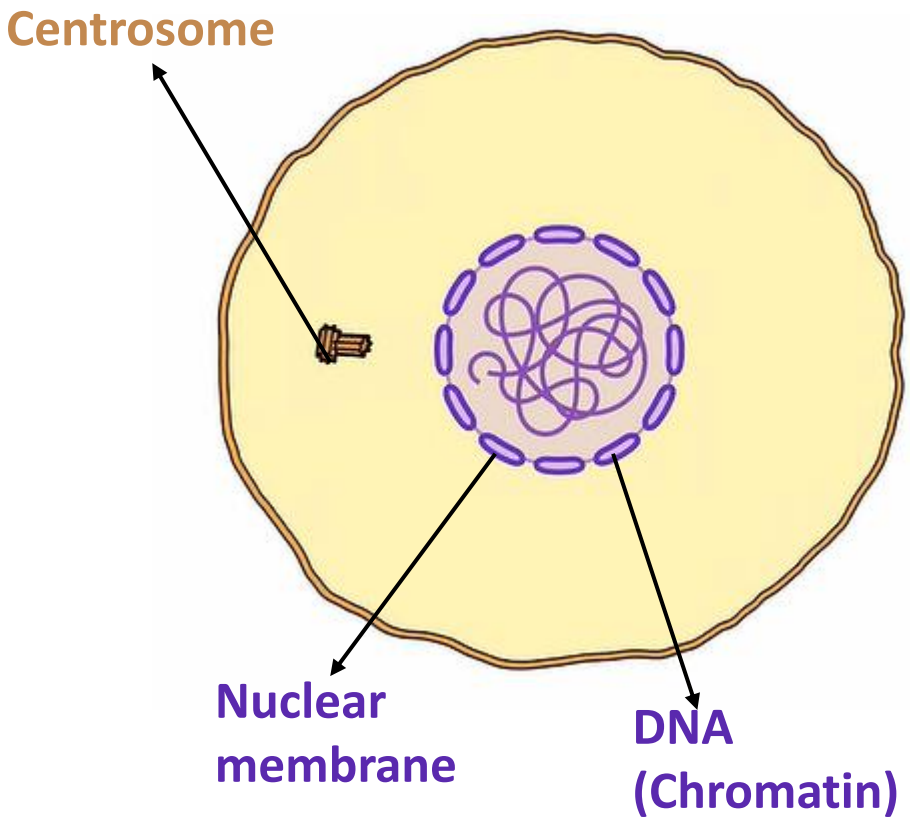
PHASES

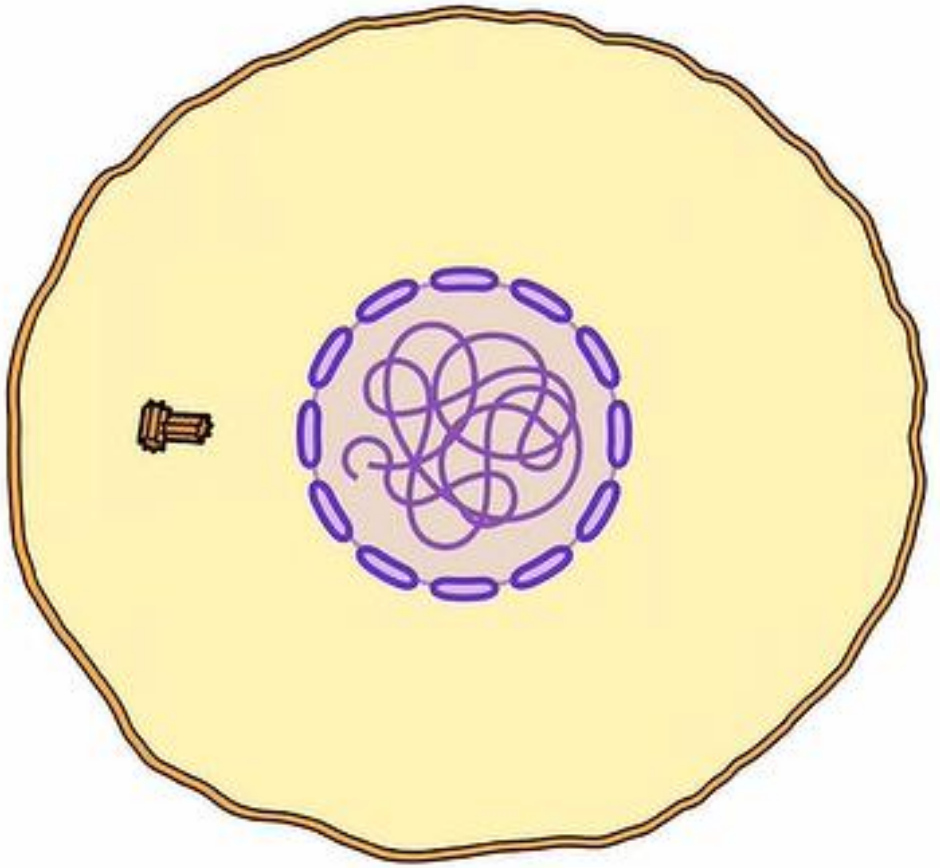


Interphase = G_1 + S + G_2

M phase = mitosis + cytokinesis

The G_1 Phase





What changes during S phase?

Before replication

- One chromosome
- One chromatid
- One Centrosome

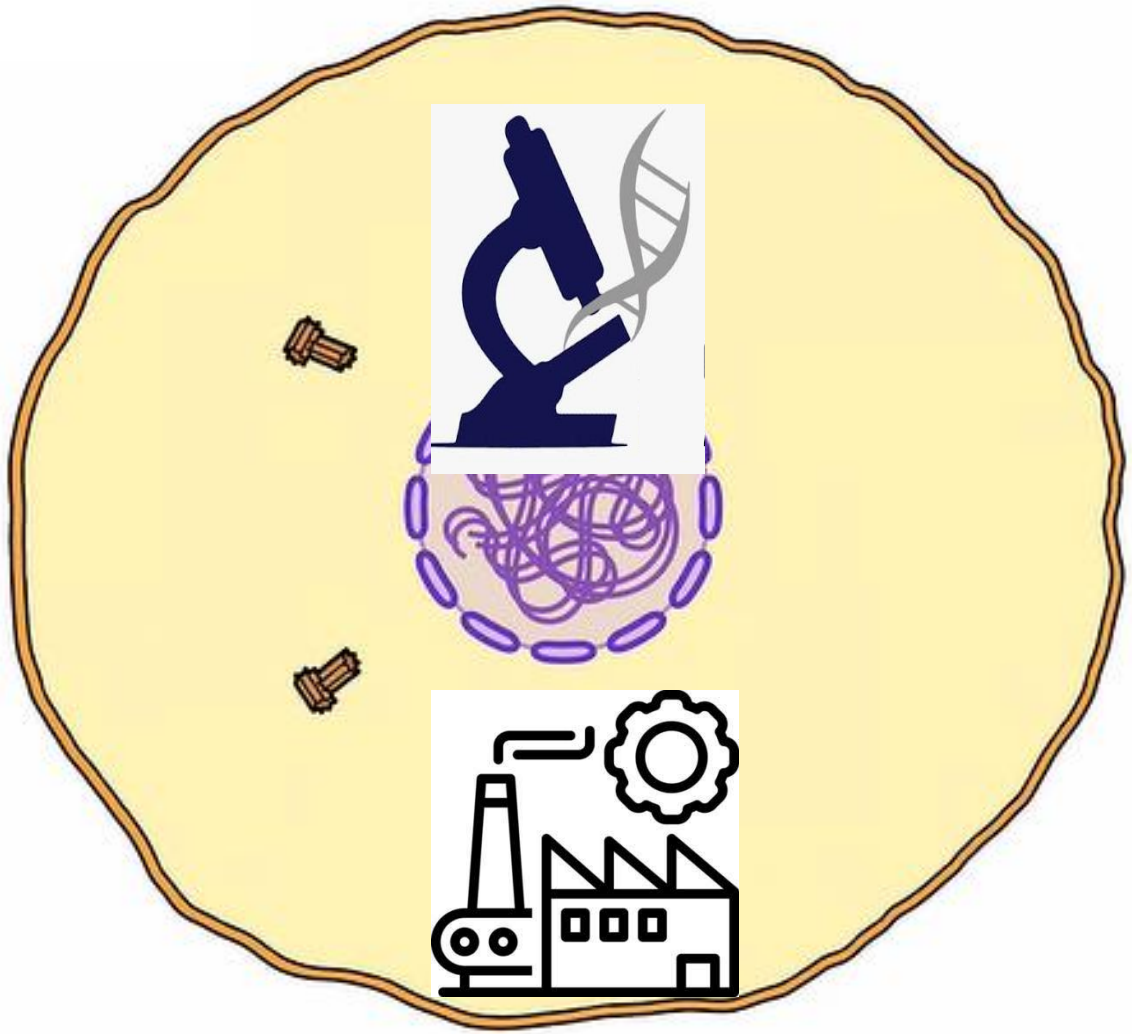
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After replication

- One replicated chromosome
- Two sister chromatids joined at the centromere
- Two Centrosome

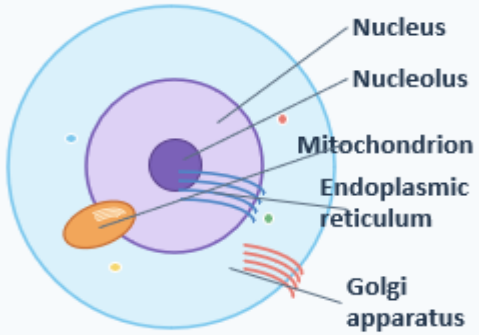
The copies must later be separated accurately.

The G_2 Phase



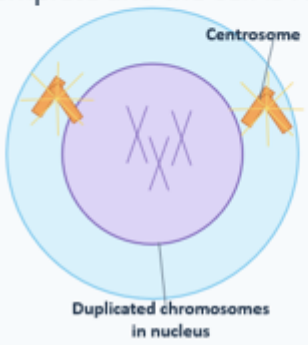
G1 Phase – Cell Growth

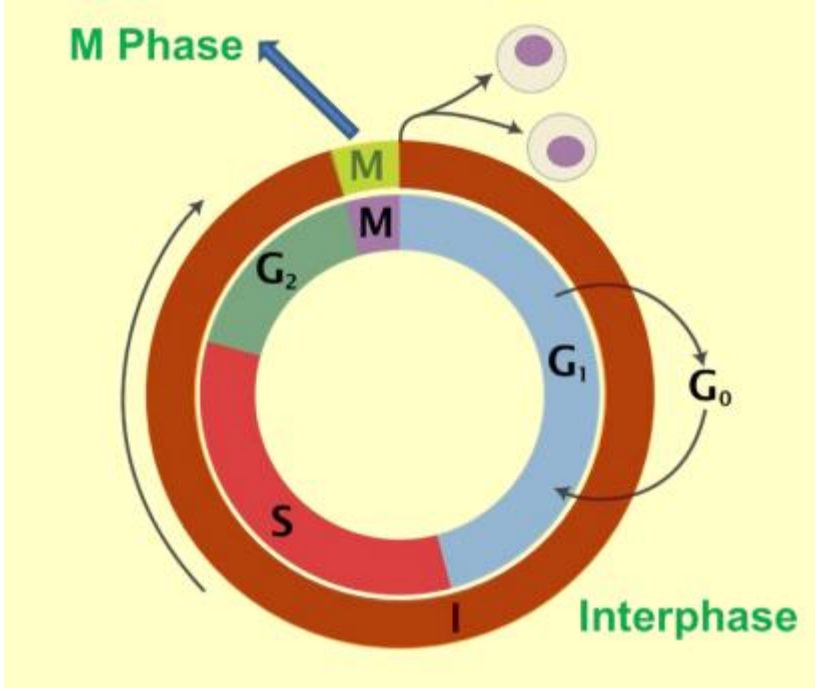
- Cell increases in size
- High metabolic activity
- Synthesis of RNA, proteins and organelles
- Monitors environmental conditions and internal signals
- Restriction point (G1 checkpoint) ensures the cell is ready to replicate DNA



G2 Phase

- Cell continues to grow
- Synthesis of proteins needed for mitosis (e.g., tubulin)
- Organelles are duplicated
- DNA is checked for errors and repairs are made
- G2 checkpoint ensures DNA replication is complete and the cell is ready for mitosis



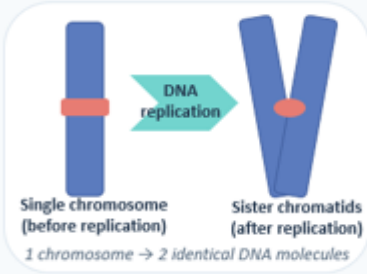


Interphase = G_1 + S + G_2

G_0 - Resting Phase

S Phase

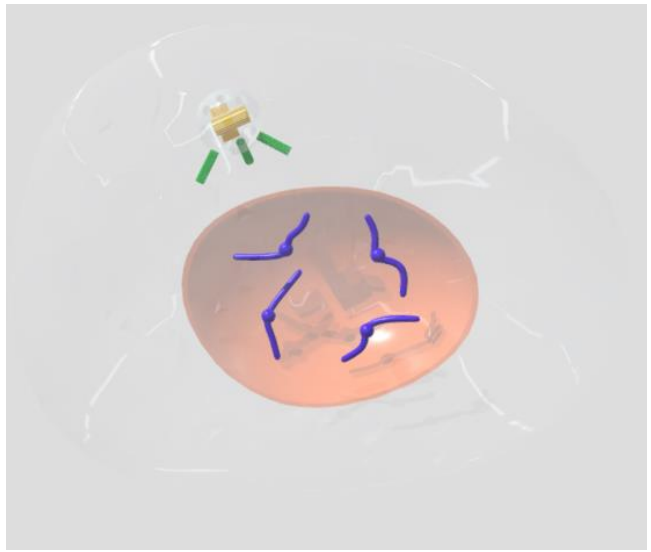
- Entire genome is replicated once and only once
- Each chromosome is duplicated to form sister chromatids
- Histone proteins are synthesized
- DNA replication is highly accurate and proofread
- Checkpoint ensures replication is complete and DNA is undamaged

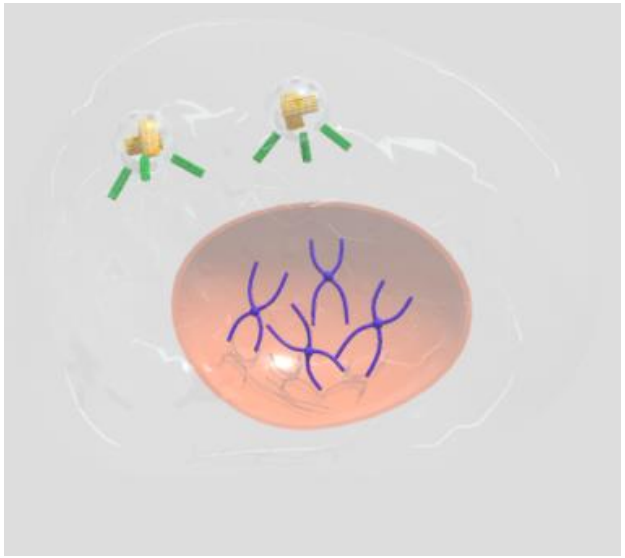


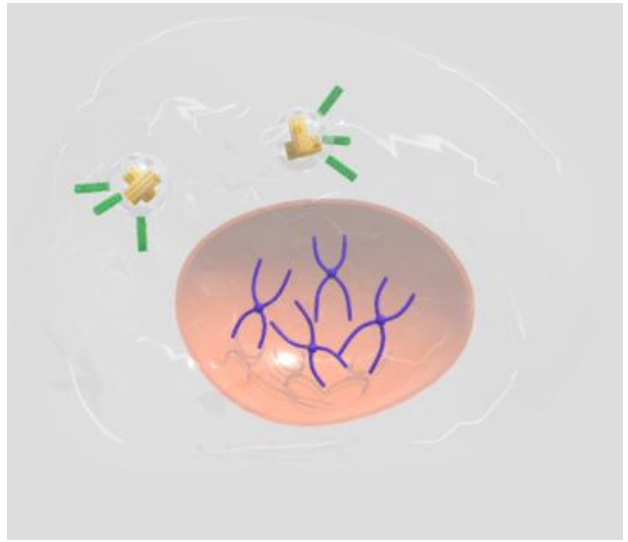
M Phase

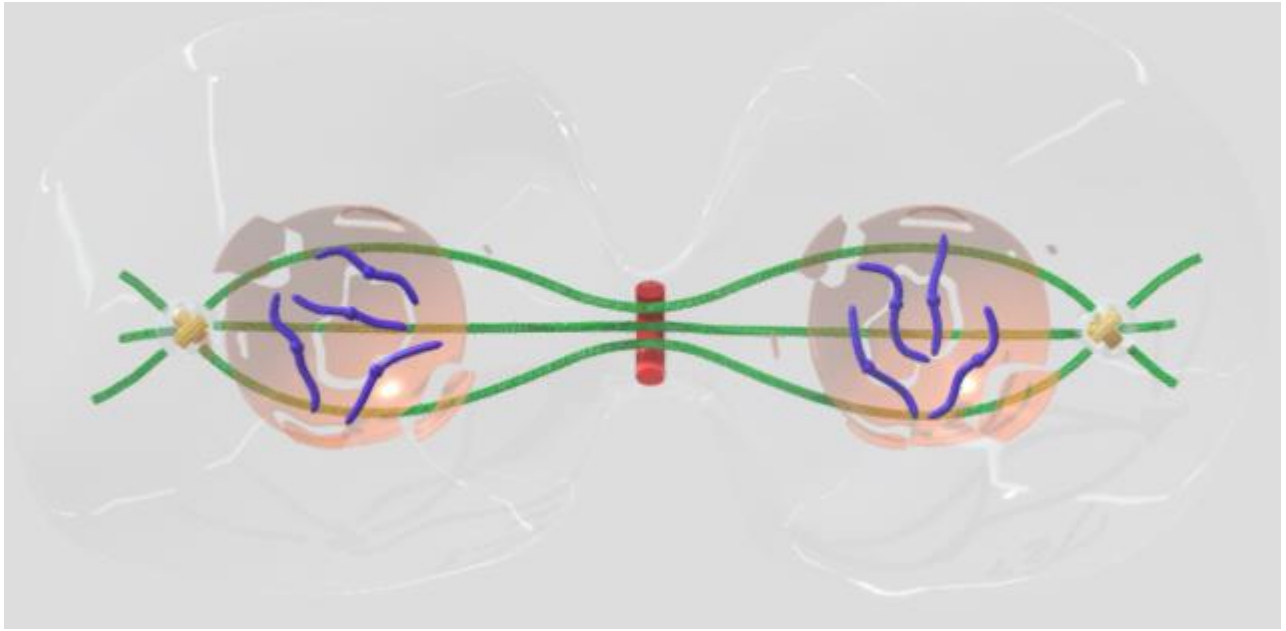
- Mitosis: division of the nucleus into two genetically identical nuclei
- Cytokinesis: division of the cytoplasm to form two daughter cells
- Ensures equal segregation of chromosomes
- Produces two genetically identical diploid daughter cells



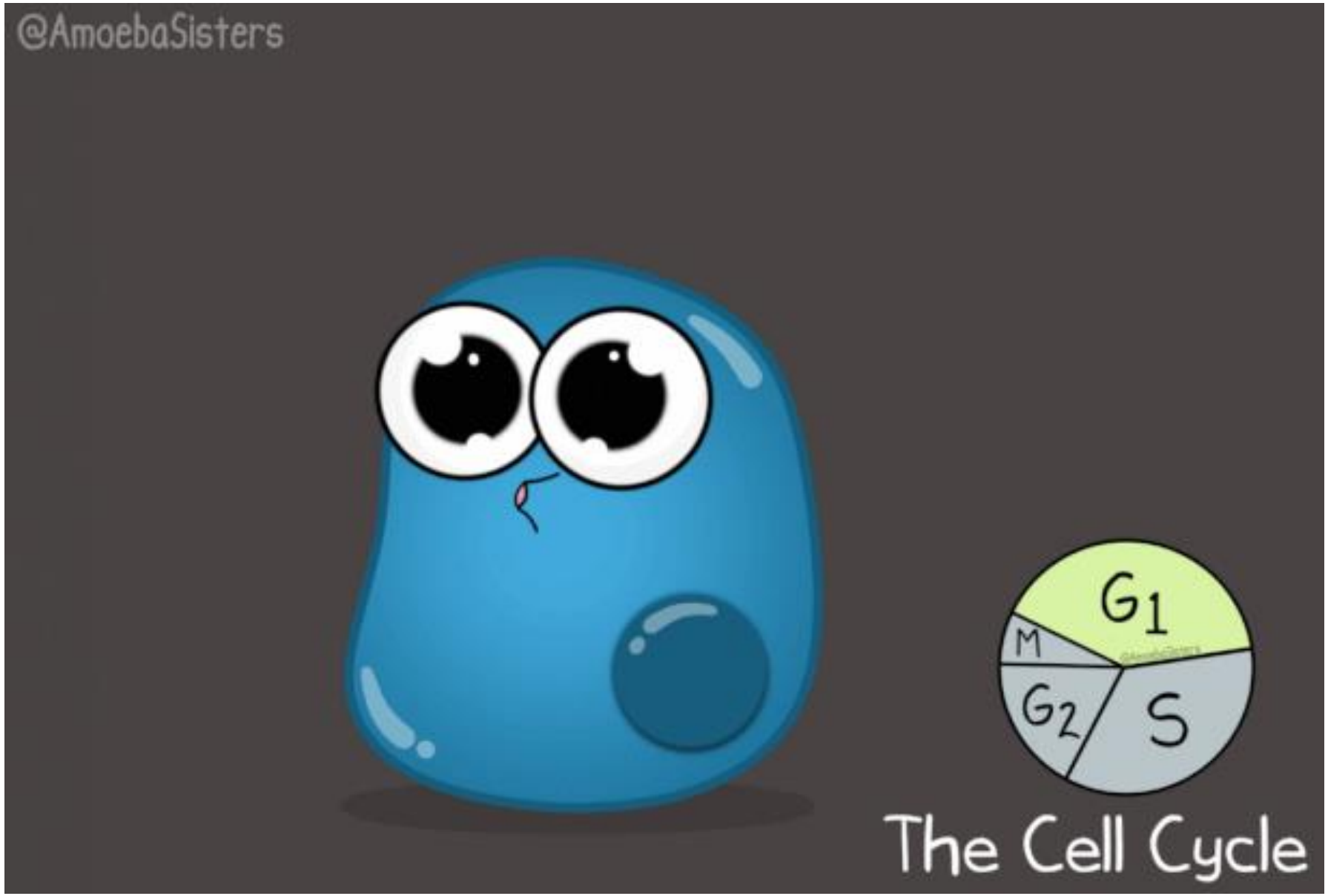








https://sketchfab.com/juliopanz/models


@AmoebaSisters
G1
M
G2
S
The Cell Cycle